Electromagnetic and Gravitational Radiation: A Parallel Development

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Abstract

In this report, we present a side-by-side derivation of electromagnetic and gravitational radiation starting from retarded potentials. For electromagnetism, the standard multipole expansion yields dipole radiation as the leading contribution. Adapting the same method to gravity by introducing retardation into the Newtonian potential, momentum conservation eliminates the dipole term. Quadrupole radiation then emerges as the dominant effect. The resulting wave amplitude and luminosity closely match full general relativity up to numerical factors of order unity. We apply these results to estimate gravitational wave luminosity in binaries and compare with solar electromagnetic output.

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1 Introduction

Radiation from time-varying sources appears in both electromagnetism and gravity, but with key differences rooted in conservation laws. While Maxwell's equations admit dipole radiation from accelerated charges, gravitational radiation requires at least quadrupole

order. This distinction follows directly from momentum conservation rather than any deep structural difference between the theories.

We develop both cases in parallel using retarded potentials. For electromagnetism, the derivation is standard [1]. For gravity, we follow an approach suggested by Schutz [2] where retardation modifies the Newtonian potential. Though this sidesteps general relativity's tensor nature, it captures the essential scaling and angular dependence while remaining accessible. Throughout, we denote field points by $\mathbf{r}$, source points by $\mathbf{r}'$, and their separation by $\mathcal{R} = |\mathbf{r} - \mathbf{r}'|$.

2 Electromagnetic dipole radiation

2.1 Retarded potentials and expansion

For localized charge and current distributions, the retarded potentials in the Lorenz gauge read

$$V(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0}\int \frac{\rho(\mathbf{r}',t_r)}{\mathcal{R}}\,d^3r', \quad \mathbf{A}(\mathbf{r},t) = \frac{\mu_0}{4\pi}\int \frac{\mathbf{J}(\mathbf{r}',t_r)}{\mathcal{R}}\,d^3r', \tag{1}$$

with $t_r = t - \mathcal{R}/c$. In the far zone ($r \gg$ source dimensions) we approximate

$$\mathcal{R} \approx r - \hat{\mathbf{r}}\cdot\mathbf{r}', \qquad 1/\mathcal{R} \approx 1/r, \tag{2}$$

giving $t_r \approx t - r/c + \hat{\mathbf{r}}\cdot\mathbf{r}'/c \equiv t' + \hat{\mathbf{r}}\cdot\mathbf{r}'/c$. Expanding the charge density,

$$\rho(\mathbf{r}',t_r) = \rho(\mathbf{r}',t') + \frac{\hat{\mathbf{r}}\cdot\mathbf{r}'}{c}\dot{\rho}(\mathbf{r}',t') + \cdots \tag{3}$$

Substituting into V,

$$V(\mathbf{r},t) \approx \frac{1}{4\pi\epsilon_0 r}\left[\int \rho\,d^3r' + \frac{1}{c}\hat{r}_i\int \dot{\rho}r_i'\,d^3r' + \cdots\right]. \tag{4}$$

2.2 Dipole term

The first integral is the total charge Q and contributes nothing to radiation. For the second, charge conservation $\dot{\rho} = -\nabla'\cdot\mathbf{J}$ yields

$$\int \dot{\rho}r_i'\,d^3r' = -\int(\nabla'\cdot\mathbf{J})r_i'\,d^3r' = \int J_i\,d^3r' = \dot{d}_i(t'), \tag{5}$$

where $\mathbf{d}(t) = \int \rho(\mathbf{r}',t)\mathbf{r}'\,d^3r'$ is the electric dipole moment. Thus

$$V_{\rm rad}(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0 cr}\,\hat{\mathbf{r}}\cdot\dot{\mathbf{d}}(t'). \tag{6}$$

The radiation fields follow from the Maxwell equations $\mathbf{E} = -\nabla V - \partial_t\mathbf{A}$ and $\mathbf{B} = \nabla\times\mathbf{A}$. In the radiation zone,

$$\mathbf{E}_{\rm rad} = \frac{\mu_0}{4\pi r}[\ddot{\mathbf{d}}]_{\rm ret}\times\hat{\mathbf{r}}\times\hat{\mathbf{r}}, \quad \mathbf{B}_{\rm rad} = \frac{1}{c}\hat{\mathbf{r}}\times\mathbf{E}_{\rm rad}. \tag{7}$$

The angular distribution of power is $dP/d\Omega = (\mu_0/16\pi^2 c)|\ddot{\mathbf{d}}|^2\sin^2\theta$, integrating to the Larmor formula

$$\boxed{P_{\rm EM} = \frac{\mu_0|\ddot{\mathbf{d}}|^2}{6\pi c}}. \tag{8}$$

3 Gravitational radiation from retarded potentials

3.1 Retarded Newtonian potential

Now, let us consider the retarded version of the Newtonian potential,

$$\phi(\mathbf{r},t) = -G\int \frac{\rho(\mathbf{r}',t_r)}{\mathcal{R}}\,d^3r', \tag{9}$$

which satisfies $\nabla^2\phi - c^{-2}\partial_t^2\phi = 4\pi G\rho$. For large r, the same expansion gives

$$\begin{aligned}\phi(\mathbf{r},t) = &-\frac{G}{r}\int \rho(\mathbf{r}',t')\,d^3r' + \frac{G}{cr}\hat{r}_i\int \dot{\rho}(\mathbf{r}',t')r_i'\,d^3r' \\ &-\frac{G}{2c^2r}\hat{r}_i\hat{r}_j\int \ddot{\rho}(\mathbf{r}',t')r_i'r_j'\,d^3r' + \cdots\end{aligned} \tag{10}$$

3.2 Emergence of quadrupole radiation

The first term is the static potential $-GM/r$. The second term involves $\int \dot{\rho}r_i'\,d^3r'$. Using mass conservation $\dot{\rho} = -\nabla'\cdot(\rho\mathbf{v})$,

$$\int \dot{\rho}r_i'\,d^3r' = -\int[\nabla'\cdot(\rho\mathbf{v})]r_i'\,d^3r' = \int \rho v_i\,d^3r' = P_i, \tag{11}$$

the total momentum. For isolated systems P_i is constant, so this term produces no radiation. This is the crucial difference from electromagnetism: **momentum conservation forbids gravitational dipole radiation**.

The first time-varying term comes from the third integral. Defining the quadrupole moment

$$I_{ij}(t) = \int \rho(\mathbf{r}',t)r_i'r_j'\,d^3r', \tag{12}$$

and noting $\int \ddot{\rho}r_i'r_j'\,d^3r' = \ddot{I}_{ij}$, we obtain

$$\delta\phi \approx -\frac{G}{2c^2r}\ddot{I}_{ij}(t')\hat{r}_i\hat{r}_j. \tag{13}$$

The dimensionless strain is conventionally $h = \delta\phi/c^2$, so

$$\boxed{h \approx -\frac{G}{2c^4r}\ddot{I}_{ij}(t')\hat{r}_i\hat{r}_j}. \tag{14}$$

3.3 Comparison with general relativity

Full general relativity gives the transverse-traceless metric perturbation

$$h_{ij}^{\mathrm{TT}} = \frac{2G}{c^4r}\ddot{I}_{ij}^{\mathrm{TT}}(t'). \tag{15}$$

While our scalar h depends on $\hat{r}_i\hat{r}_j\ddot{I}_{ij}$ and GR's h_{ij}^{TT} is a tensor with two polarizations, the functional forms match. The luminosity formulas are also close:

$$L_{\mathrm{GW}}^{\mathrm{GR}} = \frac{G}{5c^5}\langle \dddot{I}_{ij}\,\dddot{I}_{ij}\rangle, \tag{16}$$

$$L_{\mathrm{GW}}^{\mathrm{ret}} = \frac{G}{4c^5}\langle \dddot{I}_{ij}\,\dddot{I}_{ij}\rangle, \tag{17}$$

differing only by a factor 5/4. This agreement justifies using the simpler approach for estimates.

4 Astrophysical comparison: solar EM vs GW150914

4.1 Solar electromagnetic luminosity

The Sun's total electromagnetic power follows from its effective temperature and radius:

$$L_\odot = 4\pi R_\odot^2 \sigma T_\odot^4. \tag{18}$$

Adopting IAU 2015 values [5],

$$\boxed{L_\odot^{\mathrm{EM}} = 3.828 \times 10^{26}\,\mathrm{W}}. \tag{19}$$

4.2 GW150914 peak luminosity

LIGO's first detection involved a binary black hole merger [4]. Key parameters:

Parameter	Value
m_1	$36^{+5}_{-4} M_\odot$
m_2	$29^{+4}_{-4} M_\odot$
M_f	$62^{+4}_{-4} M_\odot$
Peak GW luminosity	$3.6^{+0.5}_{-0.4} \times 10^{49}\,\mathrm{W}$
Distance	$410^{+160}_{-180}\,\mathrm{Mpc}$

The peak luminosity

$$\boxed{L_{\mathrm{GW150914}}^{\mathrm{peak}} = 3.6 \times 10^{49}\,\mathrm{W}} \tag{20}$$

exceeds the Sun's electromagnetic output by

$$\frac{L_{\mathrm{GW150914}}^{\mathrm{peak}}}{L_\odot^{\mathrm{EM}}} = \frac{3.6 \times 10^{49}}{3.828 \times 10^{26}} \approx 9.4 \times 10^{22}. \tag{21}$$

To put this in perspective: at peak emission, GW150914 radiated **more power in gravitational waves than all stars in the observable universe combined emit electromagnetically**. The total energy released ($\sim 3M_\odot c^2$) dwarfs the Sun's lifetime electromagnetic output by a factor $\sim 10^4$.

5 Conclusion

Starting from retarded potentials, electromagnetic and gravitational radiation follow parallel derivations but diverge at dipole order due to momentum conservation. The resulting quadrupole formula for gravitational waves, while derived from a modified Newtonian potential, reproduces general relativity's scaling and angular dependence up to $\mathcal{O}(1)$ factors. This approach provides intuition without the full machinery of GR.

The extreme luminosity of GW150914 compared to solar electromagnetic output illustrates why gravitational wave astronomy can probe events across cosmological distances despite gravity's weakness. The factor 10^{23} difference in peak power underscores how differently these two radiations manifest in astrophysical settings.

The aim of this study is to develop the full general relativistic apparatus to demonstrate the tensor picture of gravitational wave radiation which will follow in the subsequent reports.

References

[1] D. J. Griffiths, *Introduction to Electrodynamics*, 4th ed., Pearson (2013).

[2] B. F. Schutz, "Gravitational waves on the back of an envelope," *Am. J. Phys.* **52**, 412–419 (1984).

[3] S. M. Carroll, *Spacetime and Geometry*, Cambridge University Press (2019).

[4] B. P. Abbott et al., "Observation of Gravitational Waves from a Binary Black Hole Merger," *Phys. Rev. Lett.* **116**, 061102 (2016).

[5] A. Prsa et al., "Nominal values for selected solar and planetary quantities," *Astron. J.* **152**, 41 (2016).